

EVENTFORCE MANAGEMENT SYSTEM USING SALESFORCE

College Name: KSK College Of Engineering And Technology

College Code: 8210

Team ID: SWUID-2026-379743

S.No	Student Name	Register Number	NM ID
1	Srinivasan J	821023243305	6A69C98ACAB81EA80C84B517C469A482
2	Harish V	821023243010	C69433C01C051D4AB0047213E3369A2C
3	Subikshan C	821023243306	FB9A09B511FD0538E7E4F31AACE9E4F3

TABLE OF CONTENTS:

1. INTRODUCTION

- 1.1 Project overview
- 1.2 Purpose
- 1.3 Environment and boundaries
- 1.4 Project objectives

2. IDEATION PHASE

- 2.1 Problem statement
- 2.2 User personas
- 2.3 Empathy map
- 2.4 Brainstorming and idea selection

3. REQUIREMENT ANALYSIS PHASE

- 3.1 Customer journey
- 3.2 Functional requirements
- 3.3 Nonfunctional requirements
- 3.4 Data flow

4. PROJECT DESIGN PHASE

- 4.1 Problem and solution fit
- 4.2 Data model
- 4.3 Relationship diagram
- 4.4 Solution architecture
- 4.5 Technology stack
- 4.6 Implementation choices

5. PROJECT PLANNING AND SCHEDULING

- 5.1 Product backlog and priorities
- 5.2 Iteration sequence and tracking
- 5.3 Acceptance criteria

6. DEVELOPMENT PHASE

- 6.1 Requirement analysis and planning
- 6.2 Objects and fields
- 6.3 Booking Logic and Venue Availability
- 6.4 Cancellation Approval
- 6.5 Notification Flows
- 6.6 Scheduled Apex
- 6.7 User Interface and Analytics
- 6.8 Data Migration
- 6.9 Security Configuration
- 6.10 Deployment and Maintenance

7. FUNCTIONAL AND PERFORMANCE TESTING

- 7.1 Test strategy
- 7.2 Functional test cases
- 7.3 Regression and execution evidence

8. ADVANTAGES AND LIMITATIONS

- 8.1 Advantages
- 8.2 Limitations

9. CONCLUSION

10. FUTURE SCOPE

11. APPENDIX

1 Introduction

1.1 Project overview

EventForce centralizes the event lifecycle in Salesforce. Event records connect a client and venue; an Event Vendor junction records multiple vendor assignments. Feedback links the client to the relevant event. A Lightning app provides navigation, while reports and a dashboard summarize upcoming work.

1.2 Purpose

The project replaces scattered booking details with related records, automates reminders and cancellation processing, prevents conflicting venue bookings and controls access by responsibility. It demonstrates Salesforce configuration and Apex development within a dedicated educational playground.

1.3 Environment and boundaries

The implementation uses `playful-unicorn-d2io9f-dev-ed.trailblaze.lightning.force.com`. The earlier org is outside this project environment. Sample records are labelled demonstrations or samples. No production migration has been performed; the guide treats deployment as preparation in a standalone Developer Edition environment.

1.4 Project objectives

Objective	Acceptance evidence
Centralize event operations	Six related custom objects and EventForce app navigation
Prevent booking conflicts	Venue validation and passing service tests
Make cancellations traceable	Separate approval and rejection records with history
Automate routine follow-up	Active reminder flow, queued path and daily batch schedule
Control access	Profiles, roles, permission set and saved sharing configuration

2 Ideation Phase

2.1 Problem statement

User	Problem	Expected outcome
Event coordinator	Booking information is scattered across clients, venues and vendors.	One linked event record and a consistent workflow.
Vendor manager	Event assignments and vendor information are difficult to coordinate.	Vendor records and event visibility with limited editing.
Client	Reminders and cancellation updates can be missed.	Automated notification workflows and feedback collection.
Event administrator	Conflicting bookings and informal cancellations reduce reliability.	Validation, controlled approvals and reporting.

2.2 User personas

Event Admin maintains application configuration and oversight. Event Coordinator plans events and manages cancellations. Vendor Manager maintains vendor information and views shared events. Client users provide feedback within their permitted record access. These personas describe intended responsibilities rather than measured interview findings.

2.3 Empathy map

The coordinator thinks about avoiding missed commitments, hears changing client requirements, sees venue and vendor constraints, and spends time updating schedules. The principal pains are duplicate information and booking conflicts. The desired gains are clear ownership, timely reminders and visible approval outcomes. This persona is a design assumption, not a claim of a conducted customer interview.

Perspective	Event coordinator needs
Think and feel	Avoid booking conflicts and missed commitments.
See and hear	Changing client requests, venue constraints and vendor availability.
Say and do	Plan events, confirm resources and track cancellation decisions.
Pains and gains	Repeated updates are costly; linked records and visible approvals improve coordination.

2.4 Brainstorming and idea selection

Manual registers and spreadsheets are simple to start but require repeated coordination. A separate custom portal adds development and maintenance work. Salesforce was selected because its related data model, declarative automation, Apex, access controls and reporting support the project requirements within one platform.

3 Requirement Analysis Phase

3.1 Customer journey

Client enquiry and registration, event planning, venue selection, vendor assignment, confirmation and reminder scheduling, event execution, feedback and operational reporting. A cancellation request branches into approval or rejection with a recorded final status.

Stage	System support
Enquiry and planning	Create client and event records; select a venue.
Coordination	Link vendors and validate venue/date conflicts.
Confirmation	Reserve the venue and schedule the three-day reminder.
Cancellation branch	Submit approval; record Canceled or Rejected outcome.
Closure	Collect feedback and complete past events on the daily schedule.

3.2 Functional requirements

Requirement	Implementation
Manage core information	Event, Client, Vendor, Venue and Feedback custom objects.
Assign multiple vendors	EventVendor junction with two master detail relationships.
Prevent conflicting bookings	Apex validates venue and date before saving.
Manage cancellation	Approval process with Canceled and Rejected final status updates.
Notify clients	Cancellation notification flow and three-day reminder flow.
Complete past events	Scheduled BatchCompleteEvents at 8 PM daily.
Control access	Four role/profile combinations, Feedback Manager and sharing rules.
Monitor operations	Upcoming Events by Month report and Operations Dashboard.

3.3 Nonfunctional requirements

The solution must maintain related-record integrity, restrict record visibility, process bulk records, present understandable validation messages and remain maintainable through documented metadata and tests. No measured load benchmark or production availability guarantee is claimed.

3.4 Data flow

The coordinator enters client, venue and event details. Saving the event invokes booking validation and updates venue availability. Vendor assignments connect suppliers to the event through the junction object. A confirmed event schedules the reminder; a cancellation request enters approval and its decision updates status. The daily batch completes eligible past events. Feedback, reports and the dashboard use the related records for review.

4 Project Design Phase

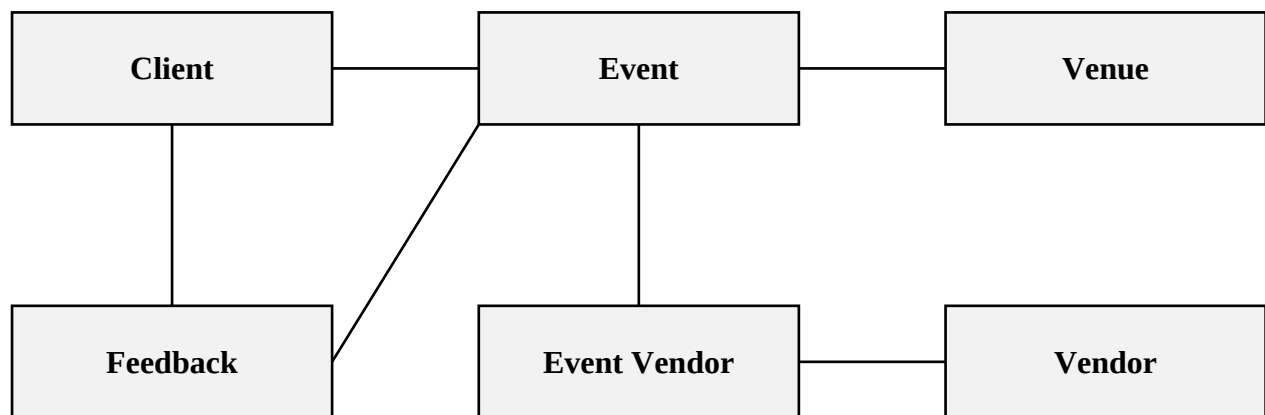
4.1 Problem and solution fit

Problem	Implemented response
Scattered client and resource details	Linked Salesforce records and a dedicated Lightning app
Conflicting venue bookings	Apex checks event date and venue before saving
Informal cancellation decisions	Approval process and recorded final status
Repeated reminder work	Scheduled flow path three days before the event
Limited operational visibility	Monthly report and operations dashboard

4.2 Data model

Object	Purpose	Relationships
Event__c	Name, type, date, status and formula budget	Lookup to Client and Venue
Client__c	Contact and location information	Referenced by Event and Feedback
Vendor__c	Services, contact and availability	Master of EventVendor
Venue__c	Address, capacity, location and availability	Referenced by Event
Feedback__c	Autonumber, rating and comments	Lookup to Client and Event
EventVendor__c	Event to vendor assignment	Master detail to Event and Vendor

4.3 Relationship diagram



Event references Client and Venue through lookups. Feedback references Client and Event through lookups. Event Vendor joins Event and Vendor through two master-detail relationships. Lines indicate related objects; see the table for relationship direction and type.

4.4 Solution architecture

Salesforce Lightning provides the user interface. Custom objects store data. Formula fields and validation rules handle declarative calculations and checks. Record-triggered flows send notifications; the approval process controls cancellation decisions. Apex handles booking conflicts, venue synchronization and scheduled completion. Reports and dashboards summarize stored data.

4.5 Technology stack

Layer	EventForce components
Platform and interface	Salesforce Trailhead Playground, Lightning app and custom tabs
Data	Six custom objects, lookups, master-detail relationships and formula fields
Automation	Record-triggered flows, cancellation approval, Apex service and scheduled batch
Access and analytics	Profiles, roles, permission set, sharing rules, report and dashboard

4.6 Implementation choices

The app is named EventForce for the requested distinct presentation. The guide demonstrates separate helper and trigger examples; this implementation consolidates venue synchronization and booking validation in EventForceService and EventForceTrigger. The report uses the deployed names so that the source code and demonstration remain traceable.

5 Project Planning and Scheduling

The implementation was organized around the guide's five phases. The following sequence is a work breakdown, not a fabricated record of team sprint velocity or individual contributions.

Phase	Work package	Completion evidence
Requirements and planning	Identify objects, users, relationships and workflows	Requirements and data model in this report
Backend configuration	Fields, formula, validation, Apex, approval and flows	Setup screenshots and Apex tests
UI customization	Tabs, app, monthly report and dashboard	Application and analytics screenshots
Data testing and security	Sample imports, profiles, roles, permissions and sharing	Import results, saved settings and test results
Deployment and maintenance	Keep source, evidence, report and demo narration	Local project files and this report

5.1 Product backlog and priorities

Priority	Work item	Outcome
High	Data model and booking integrity	Objects, relationships and validation implemented
High	Cancellation workflow	Approval and rejection demonstrated
High	Access configuration	Profiles, roles and sharing saved; session checks pending
Medium	Notifications	Flows configured; actual inbox receipt pending
Medium	Analytics and migration	Report, dashboard and sample imports verified
High for submission	Documentation and demo	Report revised; live narrated video pending

5.2 Iteration sequence and tracking

Work was sequenced into data-model setup, automation, UI and sample data, security corrections, regression testing and submission preparation. The project log and saved screenshots record observed outcomes. This report does not invent sprint dates, velocity or individual task allocations.

5.3 Acceptance criteria

Booking correctness and related data come first; notification and cancellation behavior follow. Security is checked before user demonstrations. Documentation records the implemented behavior and any remaining acceptance checks. A final submission should include the report and a screen recording narrated by the student.

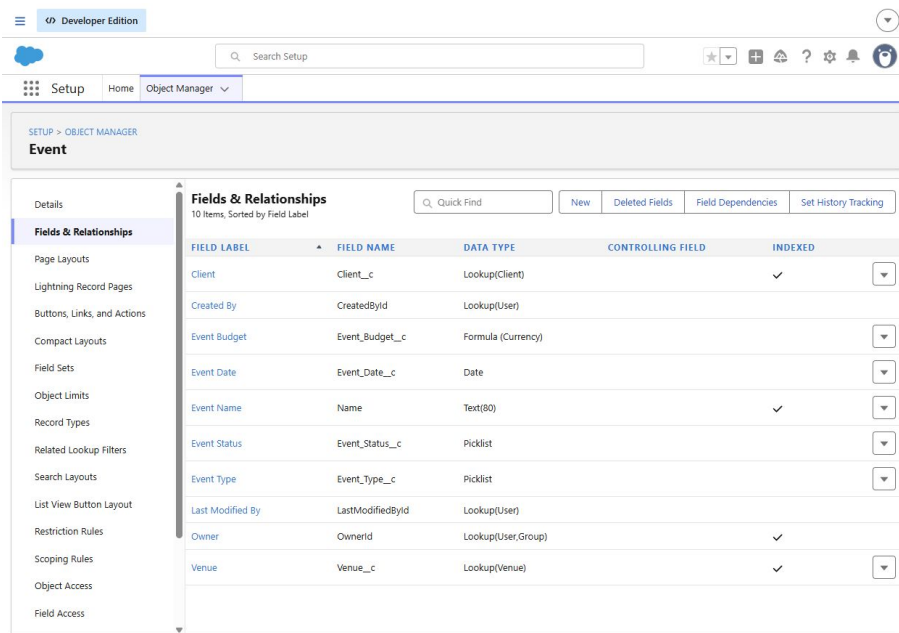
6 Development Phase

6.1 Requirement analysis and planning

The new playground separates this implementation from the earlier org. The project maps Client, Coordinator, Vendor Manager and Event Admin responsibilities to related data and controlled workflows.

6.2 Objects and fields

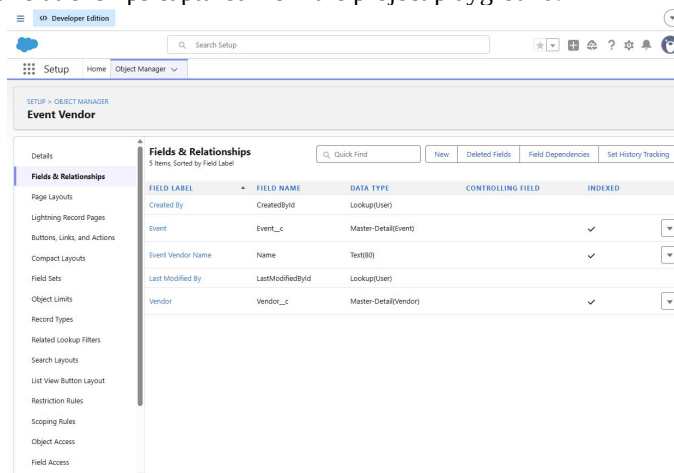
Six custom objects implement the data model. Event Type drives a Currency formula budget. Event Status tracks progression, cancellation and rejection. Client email validation and the Feedback event lookup filter protect data quality. Feedback uses an autonumber record name.



The screenshot shows the Salesforce Developer Edition interface for the 'Event' object. The 'Fields & Relationships' tab is selected, displaying a table of 10 items. The table columns are FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The items listed are Client, Created By, Event Budget, Event Date, Event Name, Event Status, Event Type, Last Modified By, Owner, and Venue.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Client	Client_c	Lookup(Client)		✓
Created By	CreatedById	Lookup(User)		
Event Budget	Event_Budget__c	Formula (Currency)		
Event Date	Event_Date__c	Date		
Event Name	Name	Text(80)		✓
Event Status	Event_Status__c	Picklist		
Event Type	Event_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Venue	Venue__c	Lookup(Venue)		✓

Figure 1. Event fields and relationships captured from the project playground.



The screenshot shows the Salesforce Developer Edition interface for the 'Event Vendor' object. The 'Fields & Relationships' tab is selected, displaying a table of 5 items. The table columns are FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The items listed are Created By, Event, Event Vendor Name, Last Modified By, and Vendor.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Event	Event__c	Master-Detail(Event)		✓
Event Vendor Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Vendor	Vendor__c	Master-Detail(Vendor)		✓

Figure 2. Event Vendor junction relationships.

6.3 Booking Logic and Venue Availability

EventForceTrigger delegates to EventForceService. The service checks venue/date conflicts across stored records and records in the same operation. Venue rows are locked during validation to serialize competing bookings. Confirmed and Pending Cancellation events require client, venue and event date.

Venue synchronization reserves venues referenced by Confirmed or Pending Cancellation events and releases them when no such event remains. Venues marked Not Available retain that state. Canceled, Rejected and Completed records do not block a new booking in this implementation.

These rules are more explicit than the guide's short example code. Planned events participate in conflict prevention, while reservation status is driven by Confirmed and Pending Cancellation events. This distinction should be explained during the demonstration.

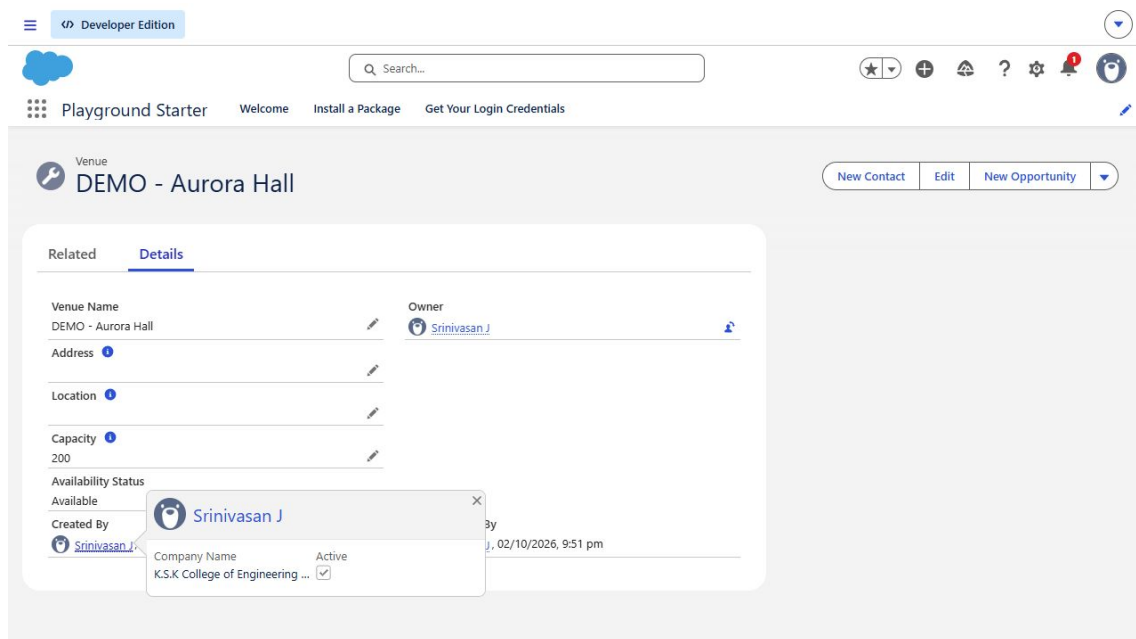


Figure 3. A venue is Available after the cancellation demonstration releases its reservation.

6.4 Cancellation Approval

The active Event Cancellation Process accepts events with Pending Cancellation status. The event owner submits the request and the configured current user acts as the demonstration approver. Submission sends an alert. Approval updates the status to Canceled; rejection updates it to Rejected, as required by the guide's field-update step.

Approval and rejection were exercised through separate demonstration records. Approval History provides the decision trail. The client cancellation flow is separate from the approval process and runs when a record changes to Canceled with a related client email.

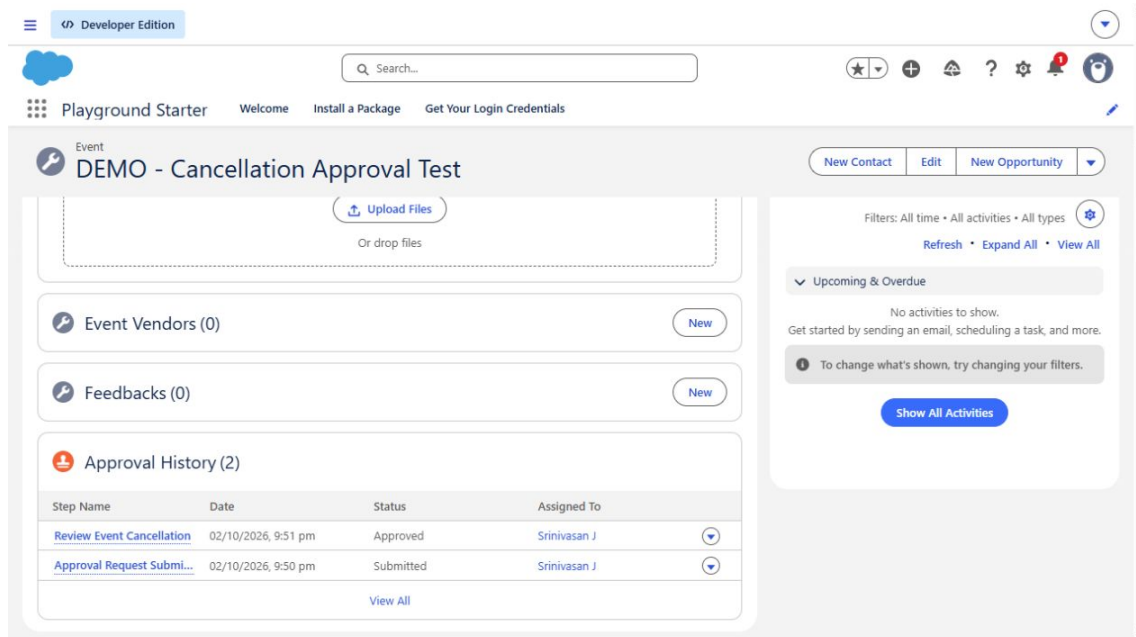


Figure 4. Completed approval history from the cancellation demonstration.

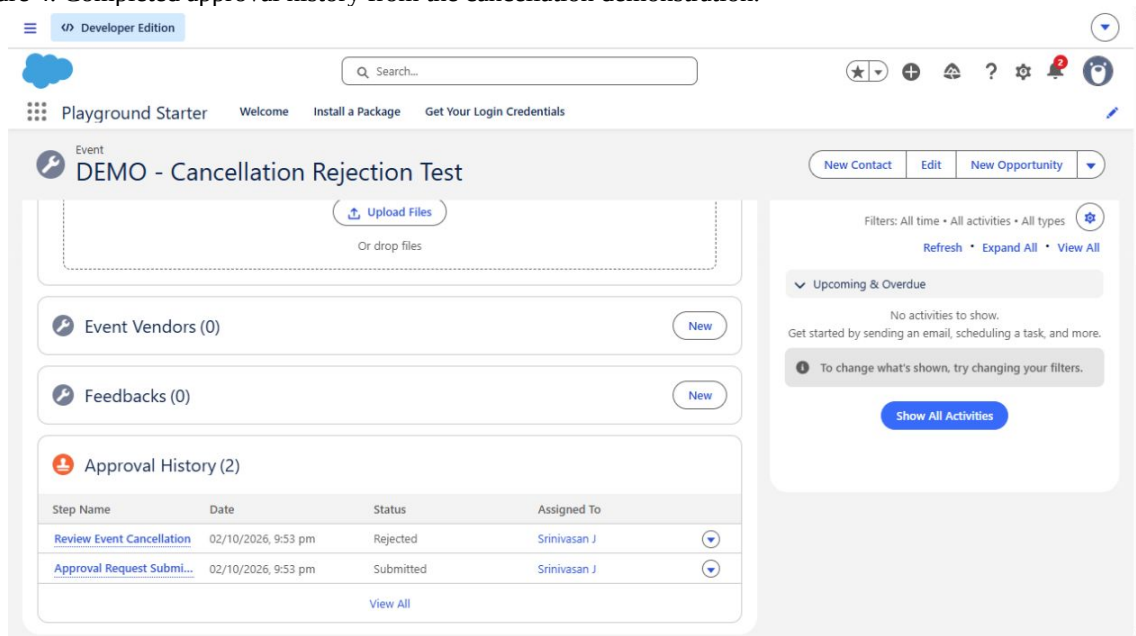


Figure 5. Completed rejection history from the rejection demonstration.

6.5 Notification Flows

Client Reminder - 3 Days Before is active. Confirmed events with date, client and email qualify. Its scheduled path uses Event Date with a three-day offset. The email goes to the client and copies the event owner. Debugging validated recipient resolution and merged event, client and venue information.

A separate cancellation notification flow sends the client an update when Event Status changes to Canceled. Entry checks avoid blank recipients and repeated notifications from unrelated edits.

The reminder demonstration for 20 October 2026 produced a scheduled path dated 17 October 2026. This verifies scheduling, not inbox delivery. Actual receipt remains a user acceptance check.

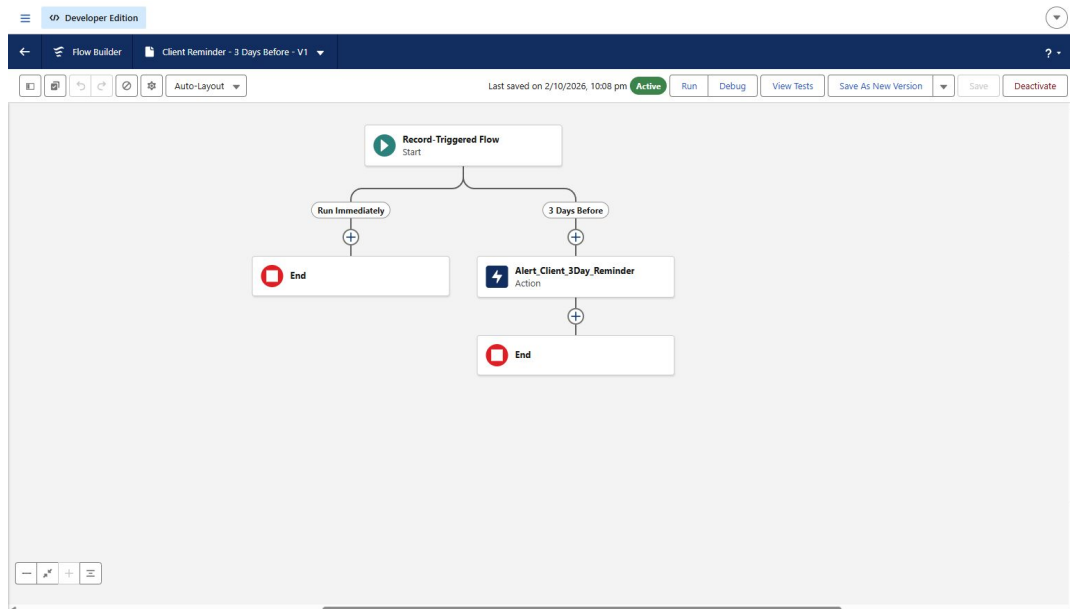


Figure 6. Active reminder flow.

Record Name	Object	Workflow Rule, Flow, or Process Name	Scheduled Date	Created By	Created Date	Automation Type
DEMO - Confirmed Reminder Test	Event	Client Reminder 3 Days Before	17/10/2026, 12:00 am	J. Srinivasan	02/10/2026, 10:11 pm	Flow Scheduled Path

Figure 7. Scheduled path evidence showing the three-day offset.

6.6 Scheduled Apex

BatchCompleteEvents implements Database.Batchable and Schedulable. To follow the guide literally, it selects events where Event Date is earlier than today and status is not Completed, then sets status to Completed. The daily job uses cron 0 0 20 * * ?, corresponding to 8 PM in the scheduling user's configured time zone.

This rule includes past Planned, Canceled and Rejected events. Consequently, a canceled event's current status can later become Completed even though its approval history remains. This is the guide's specified batch behavior and should be considered before any real production adoption.

The schedule was replaced after Salesforce required removal of the pending job before editing its class. The replacement retained the original daily time. Earlier execution on 3 October completed with zero failures and no eligible batches; automated tests separately exercise eligible records.

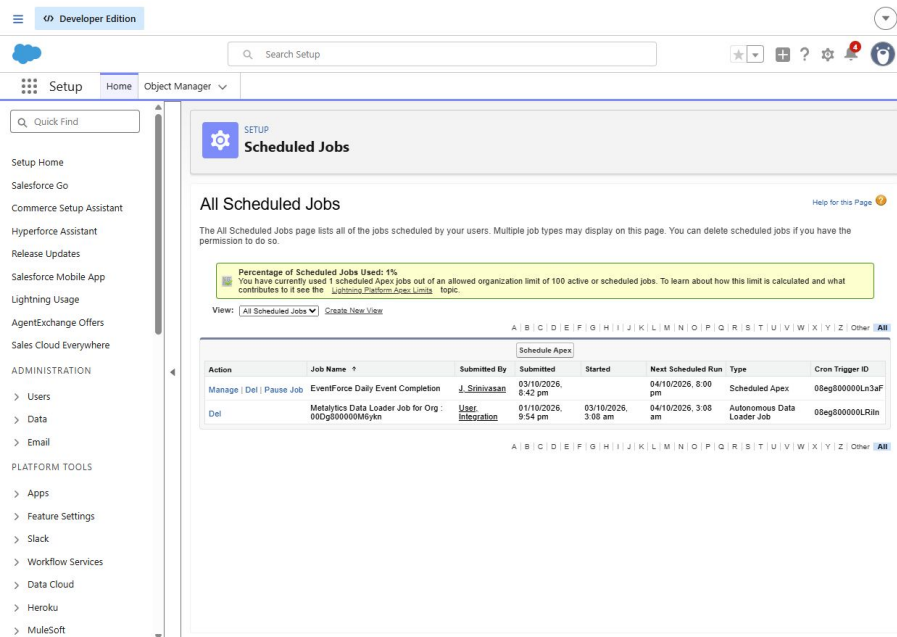


Figure 8. Restored daily schedule following the guide-aligned batch change.

6.7 User Interface and Analytics

The EventForce Lightning app includes Events, Clients, Vendors, Venues, Feedbacks, Reports and Dashboards. Five custom tabs use distinct icons. The monthly report selects all events in the next 30 days except Canceled records, groups Event Date by calendar month and summarizes record count and budget.

The EventForce Operations Dashboard uses the monthly report with a donut chart, six maximum displayed values, the subtitle Upcoming Events Overview and a light widget theme. Report results depend on the current date, so demonstration counts can change.

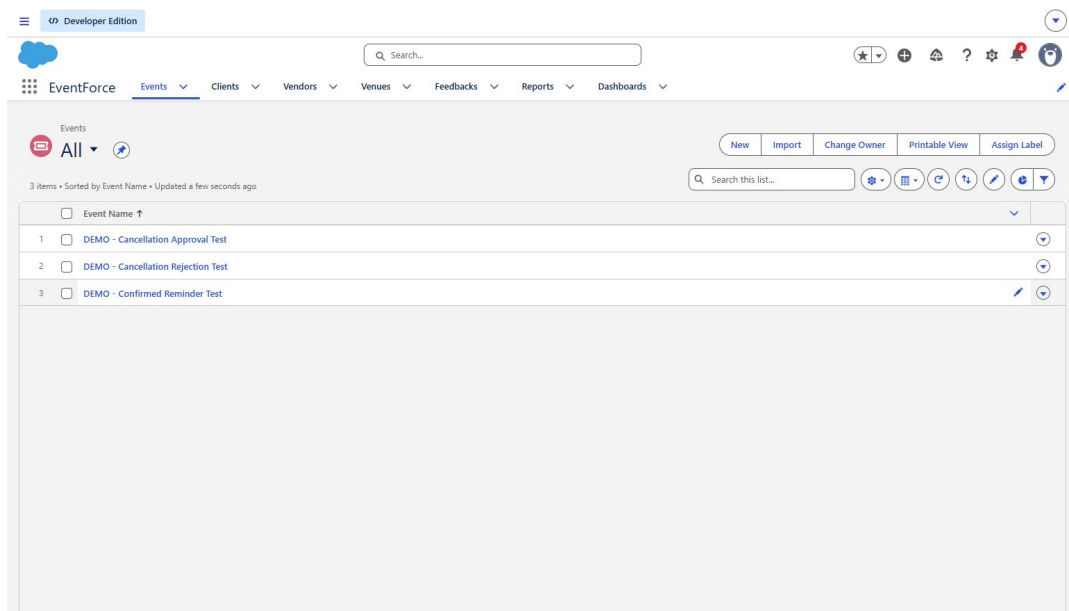


Figure 9. EventForce application and navigation.

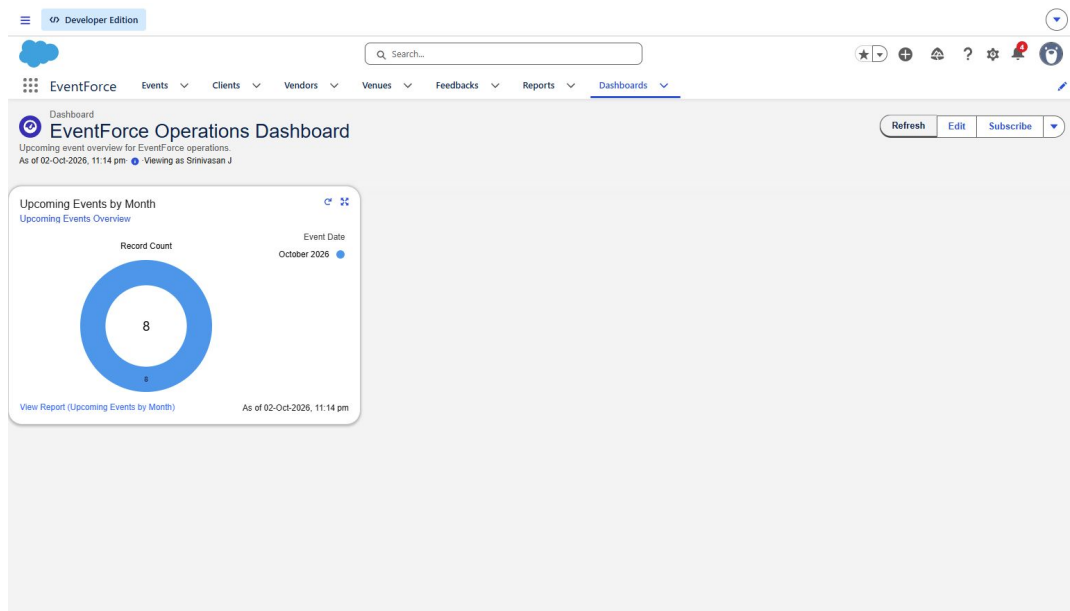


Figure 10. Operations dashboard captured during implementation.

6.8 Data Migration

Sample CSV files were prepared for all six objects. Parent records were imported before related records; relationships were matched by names. The Data Import Wizard results reported zero failed rows for all six imports. Fictional sample email addresses and labelled records distinguish demonstrations from real operational data.

Object	Records represented after import
Client	10 including the existing demo client
Venue	10 including the existing demo venue
Vendor	10
Event	13 including three demonstration events
Feedback	10
EventVendor	10

These counts describe the verified import checkpoint. Later intentional test activity may change totals. Feedback inserts should not be repeated casually because they do not have a matching duplicate key in the prepared import.

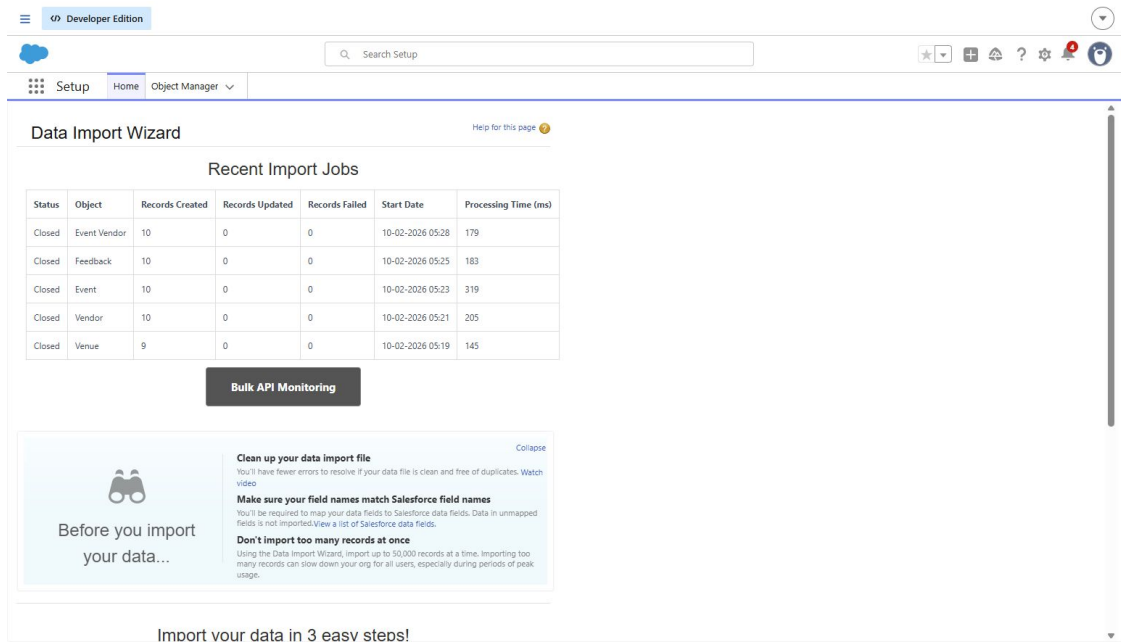


Figure 11. Data import evidence.

6.9 Security Configuration

Profile	Object permissions
Event Admin	Administrator profile clone
Event Coordinator	Event CRED; Client Read/Edit; Vendor and Venue Read; Feedback Create/Read
Vendor Manager	Vendor CRED; Event, Venue, Client and Feedback Read
Client Profile	Event Read; Feedback Create/Read; no Vendor or Venue access

CRED means Create, Read, Edit and Delete. Feedback Manager additionally grants these four Feedback permissions to the coordinator. The three operational profiles use the guide's eight-character minimum, two-hour inactivity setting and non-expiring password policy; actual role-session acceptance remains to be checked.

Event internal and external defaults are Private. Event_Sharing_For_Vendors gives the Vendor Manager role Read Only access to events owned by users in the Event Coordinator role. The report folder gives View access to those two roles. Event Admin reports to CEO; Coordinator, Vendor Manager and Client are below Event Admin.

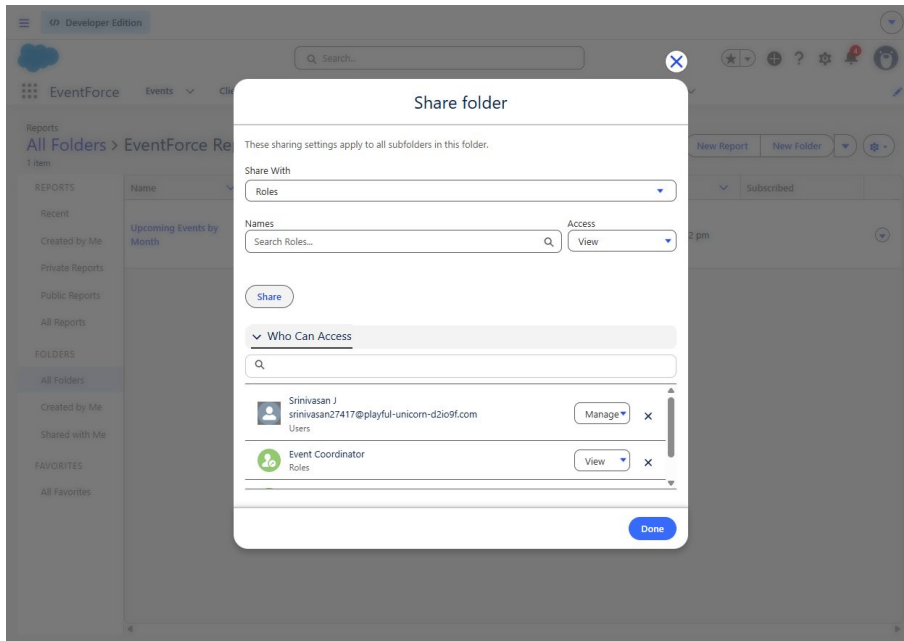


Figure 12. Saved report folder shares for the two required roles.

6.10 Deployment and Maintenance

The guide does not require deployment into a production org. The educational delivery remains in the new playground. Local metadata and sample CSVs accompany implementation evidence. They support review and future packaging, but are not represented as a fresh complete metadata export of every manually configured component.

Maintenance consists of monitoring Apex Jobs and scheduled paths, checking failed flow interviews and email delivery, reviewing access after user changes, and running regression tests after schema or Apex edits. Any future production release would require a separate environment, complete metadata retrieval and a deployment-specific test and security review.

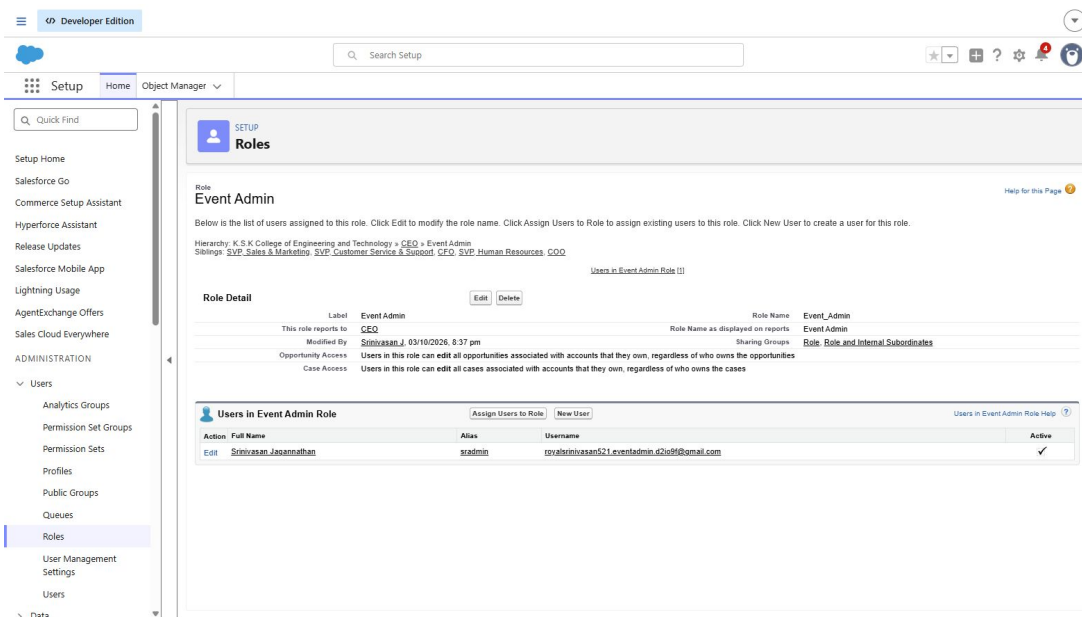


Figure 13. Saved Event Admin parent role is CEO.

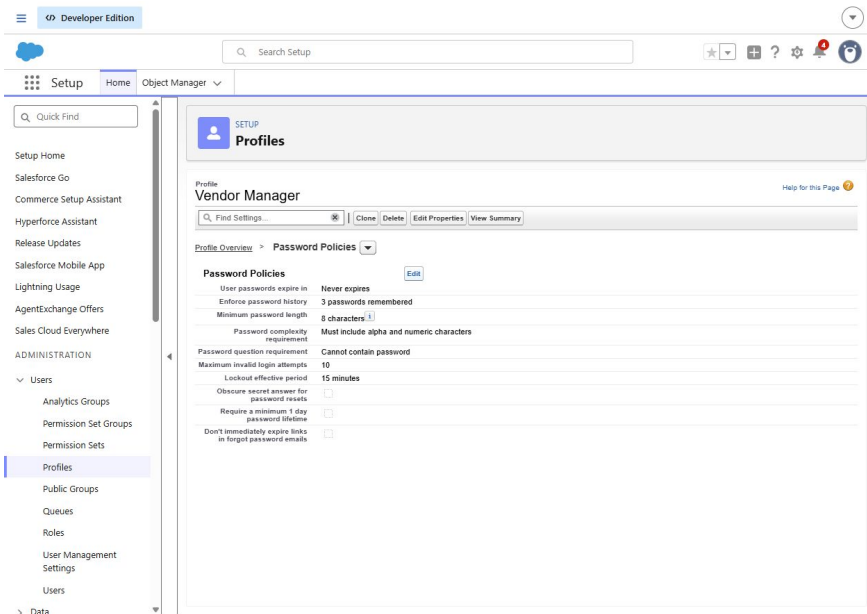


Figure 14. Vendor Manager password policy saved as specified by the guide.

7 Functional and Performance Testing

7.1 Test strategy

We used Apex tests for booking and batch behavior, separate records for approval and rejection, import-job outcomes for migration, and the scheduled-path queue for reminder timing. Configuration checks covered saved access settings. Inbox receipt, separate user sessions and load benchmarking are explicitly distinguished from the checks that passed.

7.2 Functional test cases

ID	Scenario	Expected result	Observed result
TC01	Cancellation approved	Event becomes Canceled; decision is recorded	Canceled status and Approved history verified
TC02	Cancellation rejected	Event becomes Rejected	Rejected status and history verified
TC03	Reminder for 20 October	Scheduled for 17 October	Queue showed the three-day offset
TC04	Six sample imports	No failed rows	All six jobs reported zero failures
TC05	Project regression	All six methods pass	Four service and two batch methods passed
TC06	Email receipt	Recipient receives correct message	Not verified
TC07	Role sessions	Allowed and denied actions match access model	Not verified

7.3 Regression and execution evidence

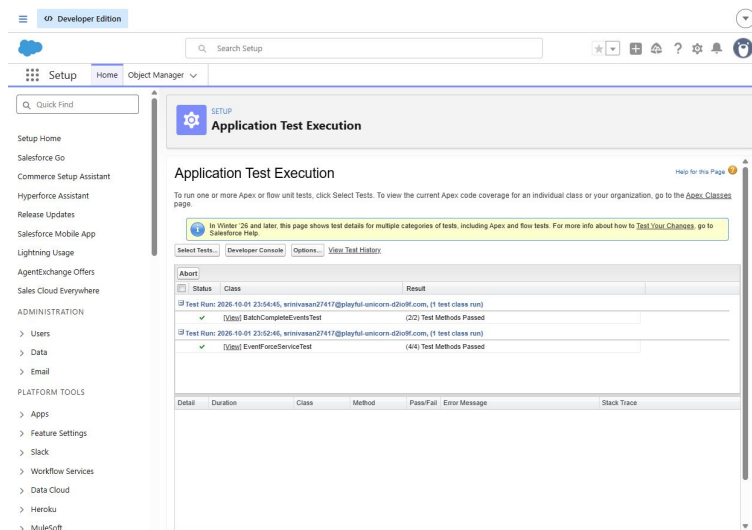


Figure 15. Earlier regression run on 1 October: four service methods and two batch methods passed. The final 3 October rerun is recorded in the project log.

Check	Observed result	Evidence or limit
Apex compilation	Successful	Invalid classes recompiled on 3 October
Booking and venue tests	4 methods passed	EventForceServiceTest
Batch tests after guide scope edit	2 methods passed	Updated regression on 3 October
Cancellation approval	Passed	Canceled status and approval history
Cancellation rejection	Passed	Rejected status and rejection history
Reminder scheduling	Passed	Three-day offset visible in queue
Sample imports	Passed	Zero failed rows in six import jobs
Scheduled execution	Completed with zero failures	No eligible records in that execution
Email inbox receipt	Not verified	A scheduled path or successful action does not prove receipt
Four user login and access checks	Not verified	Accounts and assignments verified; separate sessions not exercised
Load and concurrency benchmark	Not measured	No throughput claim

The test evidence supports the specific results shown. It is not a guarantee of zero defects. The updated batch regression passed after the guide-aligned query change.

8 Advantages and Limitations

8.1 Advantages

Related records reduce duplicate data entry and help users follow an event from client request to feedback. Automated booking checks reduce conflicting venue allocations. Approval History records cancellation decisions. Scheduled reminders and completion processing reduce repeated manual steps, while reporting makes upcoming work easier to inspect.

8.2 Limitations

The source snapshot is incomplete for clean-org deployment. Saved access configuration still requires separate user-session tests, and notification scheduling does not prove receipt. The guide-specific batch overwrites the status of past Canceled and Rejected events with Completed. The project has not been tested under production load, and no client self-service portal is claimed.

9 Conclusion

The playground implements the main EventForce workflow using Salesforce configuration and Apex. Security corrections, actual approval demonstrations, import results and regression evidence make the implementation reviewable. The remaining acceptance checks are actual email receipt and role-specific login behavior. These must not be presented as passed without observation.

10 Future Scope

Future work could add client self-service, richer vendor assignment notifications, calendar-based venue visibility, mobile-specific layouts and integrations. A production version should revisit the guide's non-expiring passwords and broad past-event completion rule. These are future improvements, not features claimed in this educational implementation.

11 APPENDIX

A. Custom Objects

Event__c
Client__c
Vendor__c
Venue__c
Feedback__c
EventVendor__c

B. Key Salesforce Records

- Events
- Clients
- Vendors
- Venues
- Feedback
- Event Vendor assignments
- Users

C. Flows

Client Reminder - 3 Days Before: record-triggered flow with a scheduled path three days before the event date.

Client Cancellation Notification: record-triggered flow for client notification when an event changes to Canceled.

D. Approval Process

Event Cancellation Process: processes cancellation requests and updates the event to Canceled on approval or Rejected on rejection.

E. Apex Components

- EventForceService - booking validation and venue availability logic
- EventForceTrigger - event trigger that invokes the service
- BatchCompleteEvents - scheduled batch for completing past events
- EventForceServiceTest and BatchCompleteEventsTest - project test classes

F. Visuals Included

- EventForce relationship diagram
- Event fields and Event Vendor relationship screenshots
- Venue availability and cancellation decision evidence
- Reminder flow and scheduled-path queue
- Daily Apex schedule
- EventForce Lightning app and operations dashboard
- Data import results
- Report sharing, role hierarchy and password-policy configuration
- Apex regression screen

THANK YOU